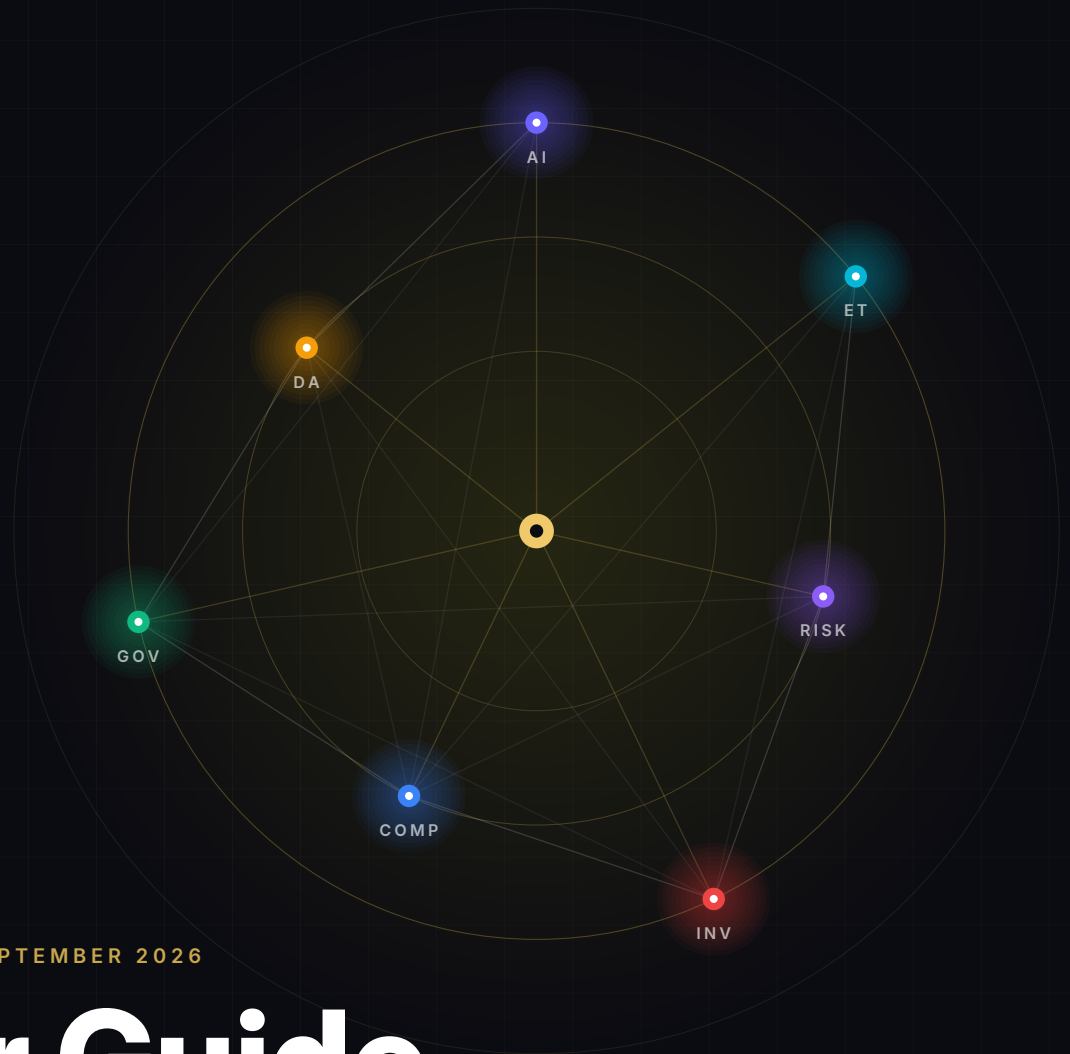




EDITION 2.0 · SEPTEMBER 2026

User Guide & Technical Review

Everything learners, partners and engineers need to know about the Chronopoli platform - illustrated end to end with a Tether (USDT) Stablecoin Infrastructure Academy.

HOSTED BY

Dubai Blockchain Center

PLATFORM

Open edX · Tutor · AWS

DISTRICTS

7 Knowledge Districts

DEMO DATA

Fictitious, for illustration

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A practical guide for learners and partners, followed by an engineering review of the platform as it stands in September 2026.

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PART ONE

Welcome to the knowledge city

What Chronopoli is, how it is organised, and the Tether demo scenario used throughout this guide.

IN THIS PART

01	Chronopoli at a glance	04
02	The Tether demo scenario	06

— CHAPTER 01

Chronopoli at a glance

Chronopoli is a global knowledge city for the digital economy. It brings regulators, law enforcement, banks, enterprises, founders and students onto one platform. Seven Knowledge Districts organise the content, and four learning layers set how deep each learner goes.

7

KNOWLEDGE DISTRICTS

One OpenEdX organisation each

4

LEARNING LAYERS

Entry to Influence

9

CUSTOM DJANGO APPS

Installed via a Tutor plugin

0

FORKS OF OPENEDX

Upstream stays upgradable

Seven Knowledge Districts

Each district is an OpenEdX organisation with its own colour, audience and course tracks. A course always belongs to one district, and partners such as Tether can publish tracks into several districts.

**AI District · CHRON-AI**

AI governance, model risk and enterprise AI for CIOs and executive teams.

**Digital Assets · CHRON-DA**

Blockchain, tokenisation, stablecoins and custody for bankers, founders and regulators.

**Governance · CHRON-GOV**

Policy, regulation and public-private dialogue for ministries and policymakers.

**Compliance · CHRON-COMP**

AML/CFT, KYC, sanctions and the Travel Rule for MLROs and compliance teams.

**Investigation · CHRON-INV**

Blockchain forensics and financial crime cases for law enforcement and FIUs.

**Risk & Trust · CHRON-RISK**

Operational, cyber and treasury risk for CROs and risk managers.

**Emerging Tech · CHRON-ET**

Quantum, IoT and spatial computing, and what comes after today's stack.

Four learning layers

LAYER	TAG	WHO IT IS FOR	TYPICAL FORMAT	PRICE BAND
L1 Entry	layer:entry	Students, newcomers, the interested public	Short self-paced courses	\$0 - \$149
L2 Professional	layer:professional	Compliance teams, executives, founders	Self-paced + verified certificate	\$299 - \$799
L3 Institutional	layer:institutional	Governments, regulators, enterprise	Advanced practitioner programmes	\$999 - \$2,499
L4 Influence	layer:influence	Thought leaders, researchers, policy shapers	Live cohorts, round tables	\$3,000 - \$8,000

How partners take part

Organisations join as **Knowledge Partners**. They host district-branded tracks and can run a white-label **Company Academy** on their own subdomain. Partners are grouped into four tiers.

TIER	EXAMPLES FROM THE PARTNER ROADMAP	WHAT THEY GET
Founding	Ripple, Cardano, Hedera	Flagship tracks, co-hosted symposia, board seat on curriculum
Strategic	Tether, Polygon, Chainlink	Branded academy, pathways and badges, talent pipeline
Content	Chainalysis, TRM Labs, Elliptic	Specialist courses inside a district
Institutional	Universities, think tanks, banks	Co-certification and research round tables

● IMPORTANT

The partner names above come from the Chronopoli partner roadmap. They do not confirm any signed agreement. All Tether figures, people, courses and screens in this booklet are **fictitious mock data** that show how the platform works.

● SCOPE

Parts I to III describe how release 2.0 is **designed to behave**. The technical review in Part IV lists the gaps found in the current code that must be closed before go-live, and the updates already applied.

— CHAPTER 02

The Tether demo scenario

This guide follows one story from start to finish: Tether opens a Stablecoin Infrastructure Academy on Chronopoli, and five learners use it. Every screen, number and name that follows belongs to this fictitious scenario.

The partner

FIELD	MOCK VALUE	WHERE IT LIVES IN THE PLATFORM
Partner name	Tether (illustrative)	<code>chronopoli_partners.Partner.name</code>
Tier	Strategic Partner	<code>Partner.tier = strategic</code>
OpenEdX org code	TETHER	<code>Partner.openedx_org_code</code>
Districts	CHRON-DA, CHRON-COMP, CHRON-INV, CHRON-RISK, CHRON-GOV	<code>Partner.districts</code> (JSON)
Academy subdomain	tether.chronopoli.io	<code>Partner.subdomain + academy_enabled</code>
Brand colours	#26A17B primary, #0F1F1A secondary	<code>brand_primary_color / brand_secondary_color</code>
Payouts	Stripe Connect acct_1TthrDEMO0000	<code>Partner.stripe_connect_account_id</code>
HR notifications	talent@tether.example	<code>Partner.hr_notification_email</code>

The people

Five learner personas and two operators take part. Their onboarding answers are fed through the real routing logic in `chronopoli_onboarding.calculate_recommendations()`. The table shows what the platform recommends for each of them.

PERSONA	ROLE (FICTITIOUS)	Q1 ROLE / Q2 FOCUS / Q3 EXPERIENCE	ROUTED TO
Layla Al-Mansoori	MLRO, Gulf Digital Exchange (VASP)	exchange / CHRON-COMP / advanced	Compliance · L2
Omar Farouk	Financial investigator, national FIU	law_enforcement / CHRON-INV / intermediate	Investigation · L2
Priya Nair	Head of Treasury Risk, regional bank	bank / CHRON-RISK / expert	Risk & Trust · L2
Mariam Saleh	Policy adviser, finance ministry	regulator / CHRON-GOV / expert	Governance · L3
Jonas Weber	Master's student, fintech	student / CHRON-DA / beginner	Digital Assets · L1
Nadia Karim	Academy administrator (Tether side)	Partner staff account	Partner dashboard
Dr. Amira Haddad	Round-table moderator, Chronopoli	Moderator role	Symposia approval

● HOW TO READ THIS GUIDE

Layla is the main character. Chapters 3 to 7 follow her from sign-up to her first badge and a job application. Chapters 8 to 12 switch to Nadia and Dr. Haddad, who run the academy and the round tables.

02

PART TWO

Learner guide

From sign-up to the first badge: onboarding, the Tether academy, payment, the AI Tutor, careers, round tables and the community.

IN THIS PART

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— CHAPTER 03

Getting started and AI onboarding

You create one account for the whole platform. A five-question onboarding then builds your personal map of the knowledge city: a home district, two neighbouring districts and the learning layer to start in.

Create your account

- 1 Open learn.chronopoli.io and choose Register**
Use your work e-mail if your employer holds a team licence. Seats are then assigned to you automatically.
- 2 Confirm your e-mail address**
Amazon SES sends the confirmation link. It is valid for 24 hours.
- 3 Start the onboarding**
After your first sign-in you are sent to /chronopoli/onboarding/. You can skip it, but your dashboard will then show generic recommendations.
- 4 Single sign-on everywhere**
The same account signs you in to the Community (Discourse), Video (Opencast) and Slides (Presenton) areas.

The five onboarding questions

#	QUESTION	ANSWER OPTIONS	LAYLA ANSWERS
Q1	Which best describes your role?	Regulator, Law Enforcement, Bank, Crypto Exchange, Enterprise, Founder, Student, Consultant	Crypto Exchange / Digital Asset Business
Q2	What is your primary area of interest?	One of the 7 districts	Compliance & AML
Q3	How would you describe your experience?	Getting started, Some experience, Experienced, Senior expert	Experienced professional
Q4	What is your primary goal on Chronopoli?	Awareness, Skills, Certificate, Policy dialogue, Network	Earn a professional certificate
Q5	How urgent is your learning need?	Now, Weeks, Months, Exploring	Within the next few weeks

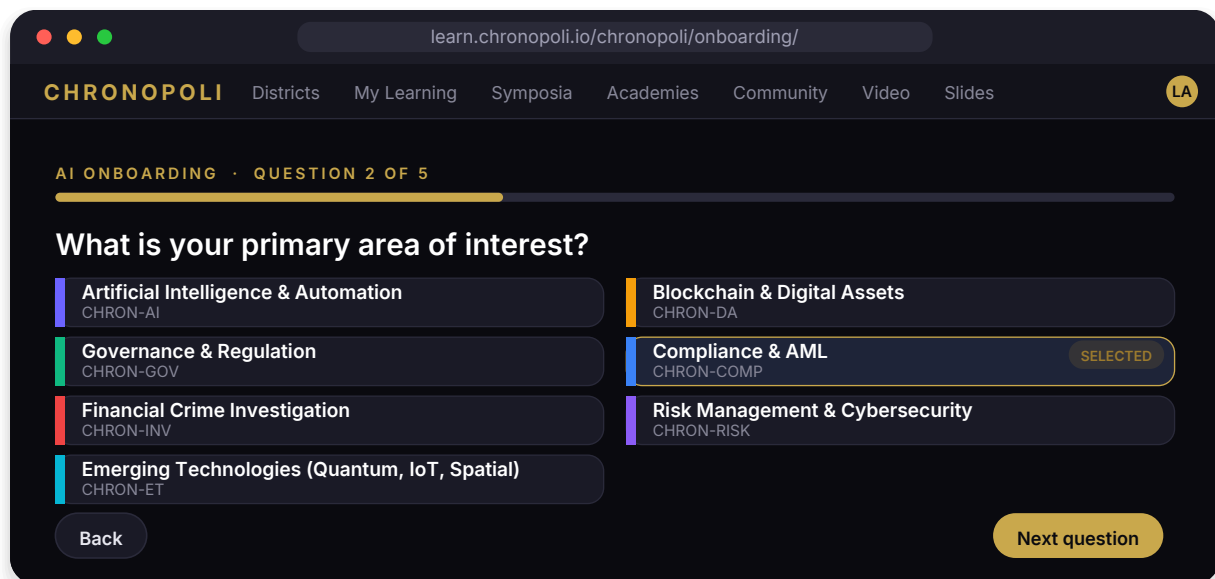


FIG. 1 Question 2 of 5. Each option shows the district colour and code. Layla picks Compliance & AML.

How the recommendation is calculated

- **Home district:** your answer to Q2. If you skip Q2, the platform uses a default for your role from Q1 (for example Regulator goes to Governance).
- **Neighbouring districts:** two complementary districts are always added. Compliance pairs with Investigation and Digital Assets.
- **Learning layer:** taken from your role. Beginners always start at L1. Advanced and expert learners never get L1.
- **Courses:** the results page lists courses from your home and neighbouring districts, ranked by layer.

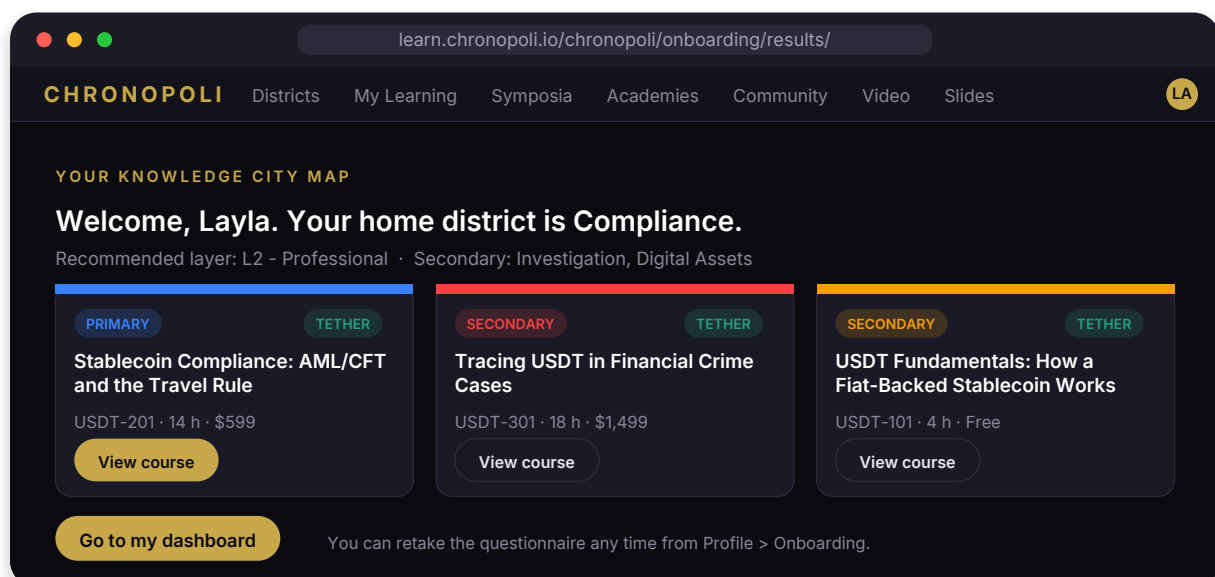


FIG. 2 Layla's result: Compliance is her home district, she starts at L2, and three Tether courses are recommended.

● NOTE

Your answers are stored in your OnboardingProfile. Other applications can read the result from </chronopoli/api/onboarding/status/>. The AI Tutor uses it to adjust its tone and the examples it picks.

CHAPTER 04

The Tether Stablecoin Academy

A Company Academy is a partner's own branded space inside Chronopoli. It groups courses into pathways, awards co-branded badges when a pathway is finished and links those badges to open roles.

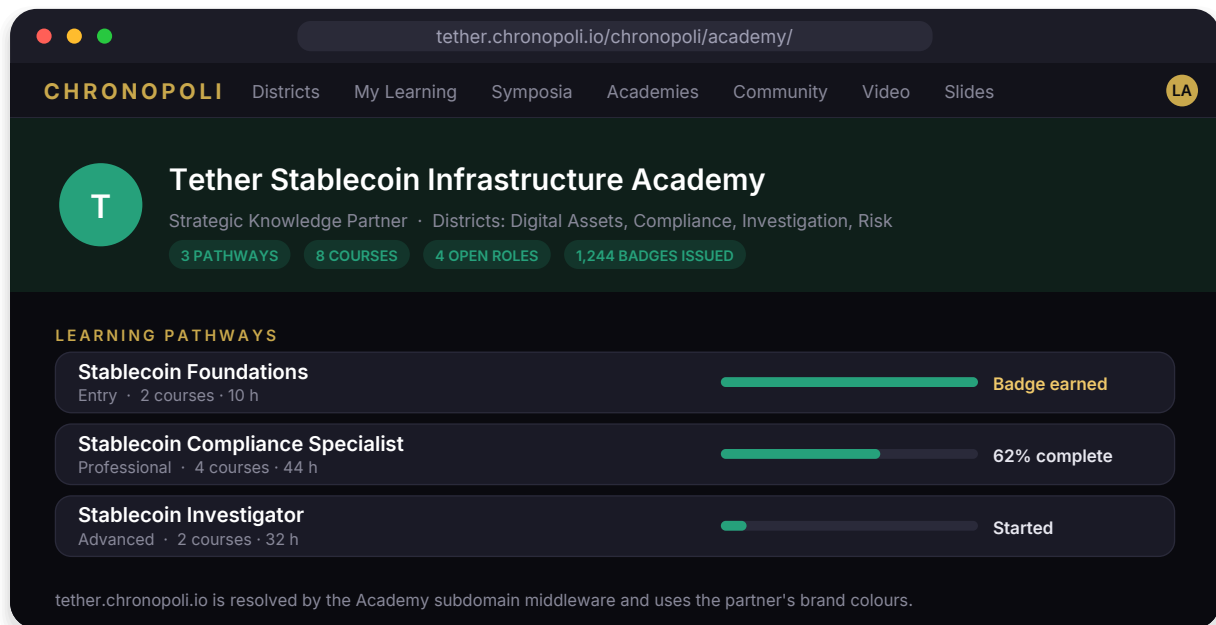


FIG. 3 The academy home page on the partner subdomain, shown with the partner's brand colours and Layla's progress.

Course catalogue (mock)

CODE	COURSE TITLE	DISTRICT	LAYER	EFFORT	PRICE
USDT-101	USDT Fundamentals: How a Fiat-Backed Stablecoin Works	DA	L1	4 h	Free
USDT-102	Multichain USDT: Issuance, Bridges and Chain Selection	DA	L1	6 h	\$149
USDT-201	Stablecoin Compliance: AML/CFT and the Travel Rule	COMP	L2	14 h	\$599
USDT-202	Reserves, Attestations and Transparency Reporting	RISK	L2	10 h	\$399
USDT-203	Stablecoin Payments and Treasury Operations	DA	L2	12 h	\$499
USDT-301	Tracing USDT in Financial Crime Investigations	INV	L3	18 h	\$1,499
USDT-302	Regulating Stablecoins: VARA, MiCA and US Frameworks	GOV	L3	16 h	\$1,299
USDT-401	Stablecoin Executive Programme (Live Cohort, Dubai)	DA	L4	5 days	\$4,500

Every course has an OpenEdX course key such as `course-v1:CHRON-COMP+USDT-201+2026_Q4`. The organisation part is the district. The partner is linked through a PartnerTrack, so a Tether course still shows up in its district's catalogue.

Learning pathways and badges

PATHWAY	LEVEL	COURSES	HOURS	BADGE AWARDED ON COMPLETION
Stablecoin Foundations	Entry	USDT-101, USDT-102	10 h	Stablecoin Foundations badge
Stablecoin Compliance Specialist	Professional	USDT-101, USDT-201, USDT-202, USDT-302	44 h	Certified Stablecoin Compliance Specialist (CSCS)
Stablecoin Investigator	Advanced	USDT-201, USDT-301	32 h	Stablecoin Investigator badge

Inside a course: USDT-201

Stablecoin Compliance: AML/CFT and the Travel Rule is Layla's first professional course. It has six modules and ends with a graded case study. Passing it with at least 70 per cent earns a verified certificate.

MODULE	TOPIC	LEARNING OUTCOME	FORMAT
1	How a fiat-backed stablecoin works	Explain issuance, redemption, reserves and why the peg matters for AML risk	Video + reading
2	Risk-based approach for VASPs	Build a stablecoin risk assessment across customers, chains and corridors	Workshop template
3	Freeze controls and sanctions screening	Know when issuer-level freezes apply and what a VASP must still do itself	Case walkthrough
4	The Travel Rule in practice	Apply FATF Recommendation 16 to USDT transfers between VASPs	Interactive lab
5	Monitoring and red flags	Spot structuring, chain-hopping and mixer exposure in stablecoin flows	Data exercise
6	Capstone case study	Write an escalation memo and file a mock suspicious transaction report	Graded assessment

CHAPTER 05

Enrolment, pricing and payment

Free courses open with one click. Paid courses use Stripe Checkout. Organisations can buy seat-based team licences, and partner revenue is split automatically.

1

Choose Enrol on a course page

Free L1 courses enrol you straight away. Paid courses open the checkout.

2

Pay with Stripe Checkout

Card, Apple Pay and Google Pay are supported. Chronopoli never stores card details.

3

Automatic enrolment

A signed Stripe webhook (checkout.session.completed) marks the purchase as paid and enrolls you in OpenEdX.

4

Receipt and revenue split

You get a receipt by e-mail. For partner courses, 70 per cent of the payment is recorded as a payout to the partner's Stripe Connect account.

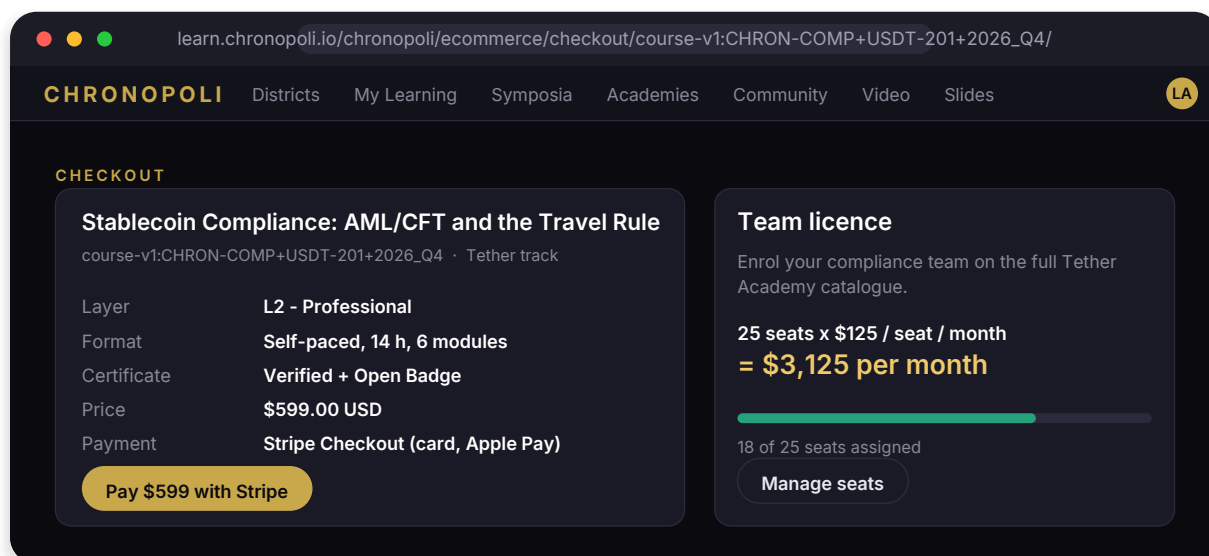


FIG. 4 Checkout for USDT-201 (left) and the team licence Layla's employer uses (right).

Team licences

Companies buy seats rather than single courses. The default list price is **\$125 per seat per month**. Seats are assigned by e-mail domain or by an administrator, and cancelled seats return to the pool. Layla's employer, Gulf Digital Exchange, holds 25 seats and has assigned 18 of them.

Revenue split on partner courses

Individual purchases of Tether courses in Q3 2026 (mock). Seats paid for through team licences are not included.

COURSE	PAID ENROLMENTS	LIST PRICE	GROSS	PARTNER 70%	PLATFORM 30%
USDT-102	380	\$149	\$56,620	\$39,634	\$16,986
USDT-201	520	\$599	\$311,480	\$218,036	\$93,444
USDT-202	260	\$399	\$103,740	\$72,618	\$31,122
USDT-203	170	\$499	\$84,830	\$59,381	\$25,449
USDT-301	140	\$1,499	\$209,860	\$146,902	\$62,958
USDT-302	90	\$1,299	\$116,910	\$81,837	\$35,073
USDT-401	50	\$4,500	\$225,000	\$157,500	\$67,500
Total	1,610		\$1,108,440	\$775,908	\$332,532

● NOTE

Cancelled checkouts leave no enrolment and no partner payout. Every Stripe event is stored once, keyed by its event ID, so a webhook that Stripe sends twice cannot enrol a learner twice.

CHAPTER 06

Learning with the AI Tutor

The AI Tutor is a chat assistant built into every course page. It answers from the Chronopoli knowledge base and shows the lessons it drew on, so you can check every answer.

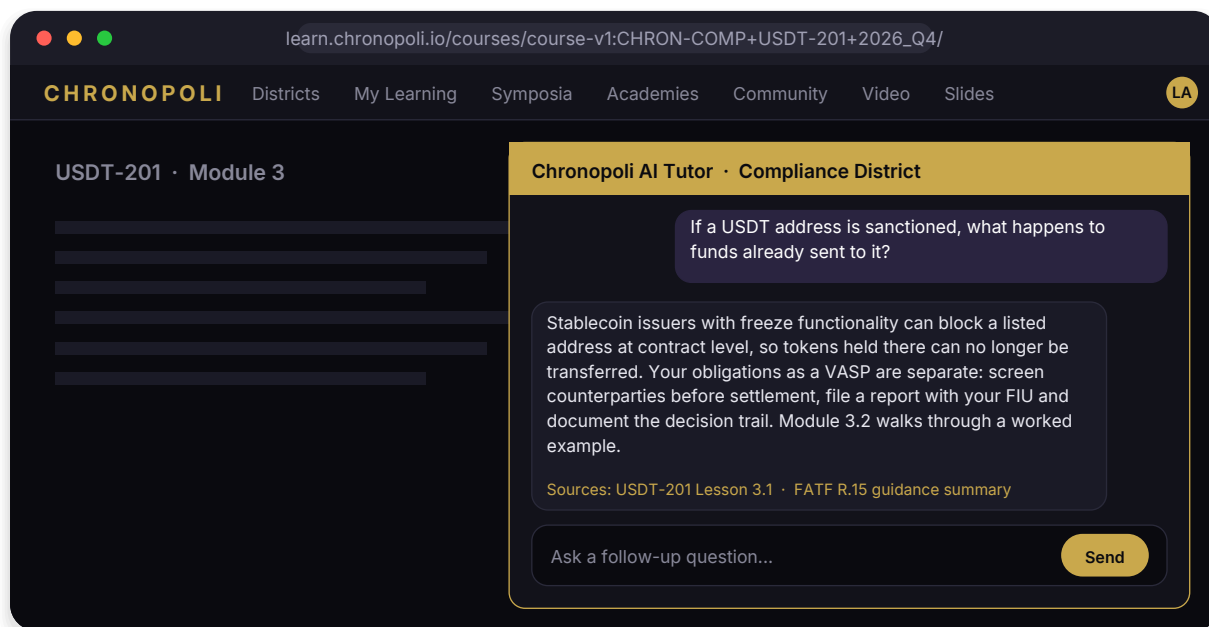


FIG. 5 The AI Tutor widget during module 3 of USDT-201. Each answer cites the course material it used.

What the tutor knows about you

- Your **home district and layer** from onboarding. This sets the depth and the examples it uses.
- The **course and unit** you are viewing, so short questions such as "what does this mean?" make sense.
- Your earlier **tutor sessions**. You can reopen them under History.

Getting good answers

INSTEAD OF ...	TRY ...	WHY
"Explain stablecoins"	"Explain how USDT redemption works for an exchange that is not a direct Tether customer"	A specific role and scenario gives a specific answer
"Is this allowed?"	"Under the VARA rulebook, can a VASP accept a USDT transfer with incomplete Travel Rule data?"	Naming the rulebook makes the tutor cite sources
"Quiz me"	"Give me five scenario questions on module 3 at L2 level, then mark my answers"	Structured practice that fits your layer

● IMPORTANT

The AI Tutor is a learning aid. It does not give legal, investment or compliance advice. Always check regulatory questions against the primary source that the tutor cites.

CHAPTER 07

Badges, careers and the talent pipeline

When you finish a pathway you receive a verifiable Open Badge. Partners can link open roles to badges, so your credentials go straight to the teams that are hiring.

Your badge

Layla finishes the **Stablecoin Compliance Specialist** pathway and receives the **Certified Stablecoin Compliance Specialist (CSCS)** badge. It follows the Open Badges 2.0 standard, which LinkedIn, Credly-compatible wallets and HR systems can all read.

OPEN BADGES 2.0 ASSERTION (MOCK)

```
{
  "@context": "https://w3id.org/openbadges/v2",
  "type": "Assertion",
  "id": "https://chronopoli.io/chronopoli/academy/api/assertions/5b0e...c41",
  "recipient": { "type": "email", "hashed": true, "identity": "sha256$9f2c..." },
  "badge": "https://chronopoli.io/chronopoli/academy/badge/cscs-tether/",
  "issuedOn": "2026-09-18T10:42:00Z",
  "verification": { "type": "hosted" }
}
```

Open roles linked to badges (mock)

ROLE	LOCATION	REQUIRED BADGE	APPLICANTS
Stablecoin Compliance Analyst	Dubai, hybrid	CSCS	21
On-chain Investigator	Abu Dhabi	Stablecoin Investigator	9
Treasury Operations Associate	Remote (GST)	Stablecoin Foundations	5
Partnerships Manager, MENA	Dubai	Any Tether badge	2

1

Opt in

In Profile > Careers, allow partners to see your badges. You can withdraw at any time. (The consent gate is a go-live requirement, see finding R-37 in Part IV.)

2

Automatic matching

When a badge is issued, the talent pipeline checks it against active roles that require that badge.

3

One-click or automatic application

A snapshot of your credentials is submitted and the partner's HR contact is notified by e-mail.

CHAPTER 08

Symposia round tables

Round tables are closed expert sessions, streamed live and held under the Chatham House Rule. After each session, an AI pipeline turns the recording into a transcript, an analysis and a set of publications. A person approves every output before it goes out.

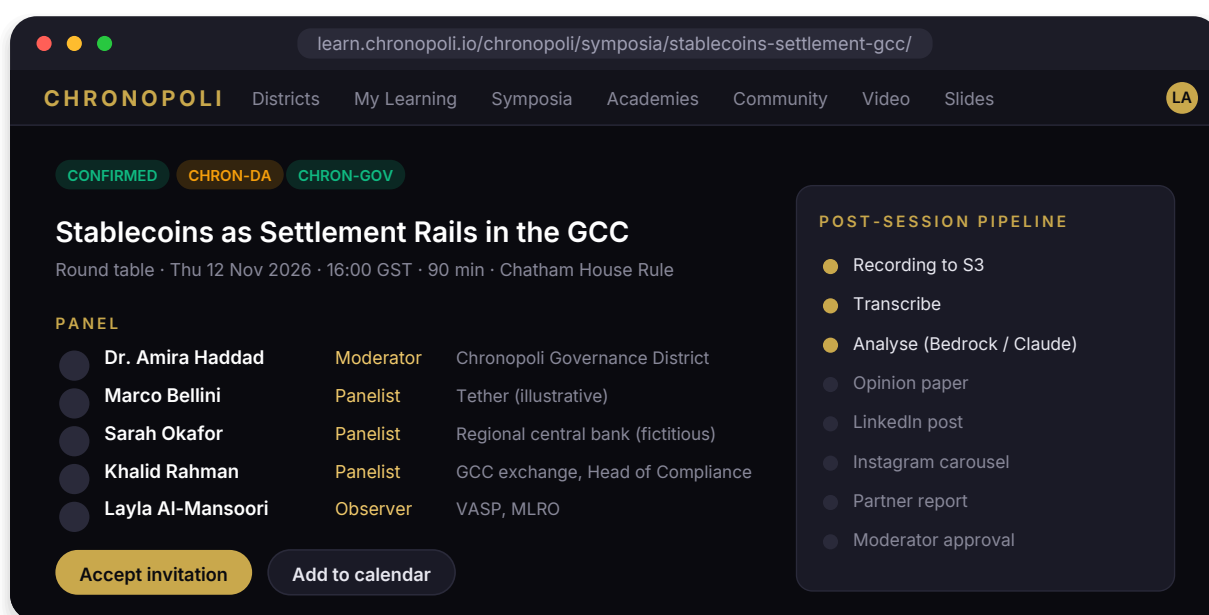


FIG. 6 A confirmed round table with panel roles on the left and the post-session pipeline on the right.

Lifecycle of a round table

STATUS	WHAT HAPPENS	WHO ACTS
Draft	Title, districts, agenda and date are prepared	Initiator
Open	Invitations go out with a secure token and an expiry date	Moderator
Confirmed	Enough panellists have accepted; an AWS IVS channel is created	Platform
Live / Recording	The session is streamed and recorded to S3	Panel
Processing	Step Functions runs transcription, analysis and the output generators	Pipeline
Completed	Every output is waiting for review in the approval queue	Moderator
Published	Approved outputs are released to learners, partners and social channels	Moderator

What the pipeline produces

Transcript

Full text from Amazon Transcribe, with speakers separated.

Analysis

Key themes, points of agreement and dissent, and open questions.

Opinion paper

A PDF white paper in Chronopoli design, with no names attached (Chatham House).

Partner report

A briefing for the sponsoring partner, here Tether.

LinkedIn post

A ready-to-post summary for the district's channel.

Instagram carousel

Slide texts and a visual brief for a carousel.

● KEY POINT

Nothing is published automatically. Dr. Haddad reviews each output in `/chronopoli/symposia/approve/` and can approve it, edit it or reject it.

— CHAPTER 09

Community, video and slides

Three companion services extend the LMS. You reach them from the top navigation and sign in with your Chronopoli account.

AREA	ADDRESS	WHAT YOU DO THERE	BUILT ON
Community	<code>community.chronopoli.io</code>	Discuss in district groups and join private cohort spaces	Discourse + Chronopoli SSO
Video	<code>video.chronopoli.io</code>	Watch lecture recordings and round-table replays	Opencast
Slides	<code>slides.chronopoli.io</code>	Instructors generate district-branded slide decks with AI	Presenton

District groups in the community

When you first sign in to the community, the SSO handshake adds you to the group for your home district. Layla joins **compliance-district** and can see the Tether Academy category **#stablecoin-compliance**. Moderators and partner staff get extra groups from their roles.

Slides for instructors

Presenton uses per-district prompt templates. The template sets colours, tone and slide structure. A Tether instructor can ask for "a 12-slide L2 lesson on Travel Rule data fields for USDT transfers" and receives a deck in Compliance District design that can be edited further.

03

PART THREE

Partner & admin guide

How Tether runs its academy: set-up, dashboards, Digital Twin masterclasses and a quick reference for operators.

IN THIS PART

10	Running a partner academy	23
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— CHAPTER 10

Running a partner academy

This chapter is for partner administrators like Nadia. It explains how to set up an academy, publish pathways and get paid. Most steps happen in the Django admin at /admin/.

1

Create the Partner record

Enter the name, tier, org code, districts and contact. Chronopoli staff do this during partner onboarding.

2

Enable the academy

Set a subdomain (tether), brand colours and a logo, then set `academy_enabled`. The subdomain middleware serves the academy on `tether.chronopoli.io`.

3

Connect payouts

Add the Stripe Connect account ID. Payouts are created when a purchase is completed.

4

Publish tracks and courses

Build the course in Studio under the district organisation, then link it with a `PartnerTrack`.

5

Build pathways

Create a `LearningPathway`, add ordered `PathwaySteps` (course keys), mark steps as required and publish.

6

Define badges

Attach a `Badge` to the pathway with its criteria, image and issuer. It is issued automatically when the last required step is complete.

7

Post roles

Create `PartnerJobPostings` with the required badges and an expiry date. Applications arrive in the admin and by e-mail.

● TIP

Brand colours only change the partner's academy. The rest of the platform keeps the Chronopoli dark and gold design, so every academy still looks like part of the same city.

CHAPTER 11

Partner dashboard and intelligence

Partners see how their academy is doing in real time: enrolments, completions, badges, revenue and reports built from round tables and learner questions.

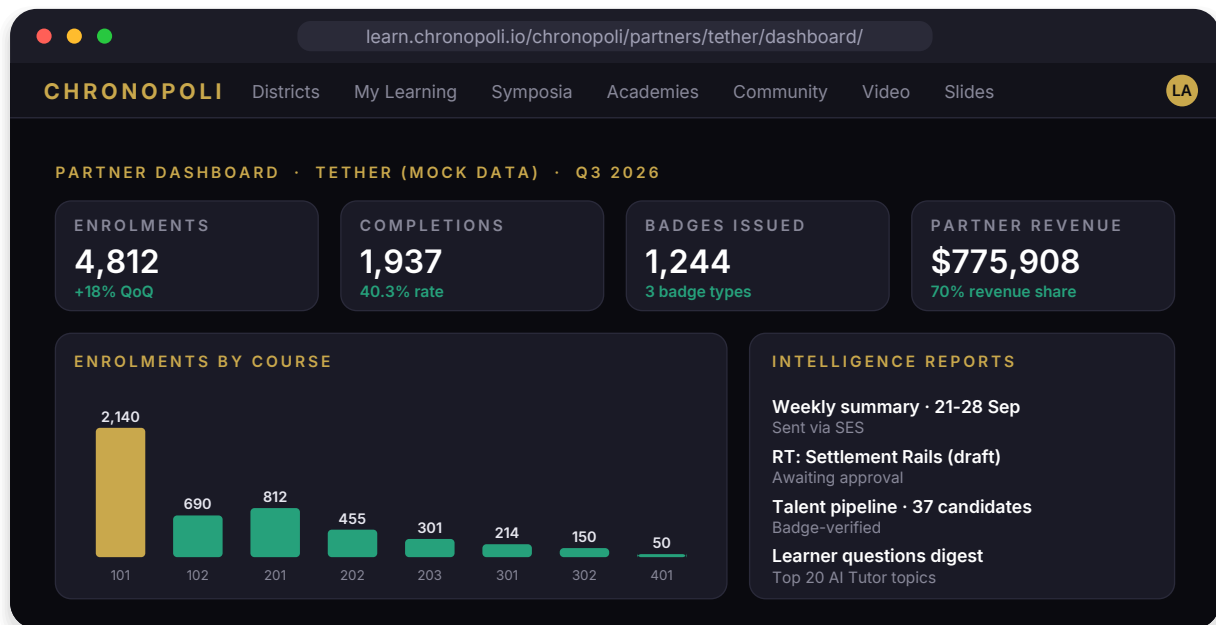
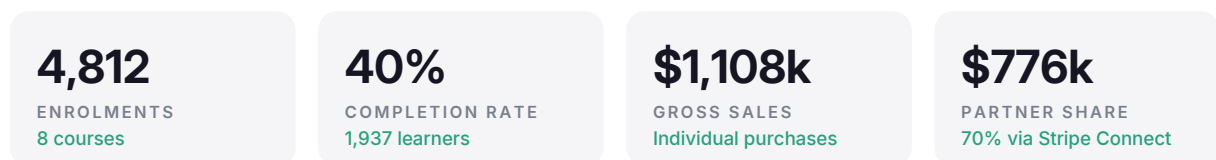


FIG. 7 The Tether partner dashboard (mock data, Q3 2026).



The weekly report

Once a week, a scheduled Celery task builds the partner report, and Amazon SES e-mails the summary to the partner contact: new enrolments, completions, badges, job applications, active tracks and revenue for the past seven days. The same data is available as JSON from </chronopoli/partners/api/tether/analytics/>.

CHAPTER 12

Digital Twin masterclasses

A Digital Twin is an AI version of a real expert. It is trained on their published work, and they approve it before it can teach. It prepares interview questions, speaks in a cloned voice and can appear as an avatar in live masterclasses.

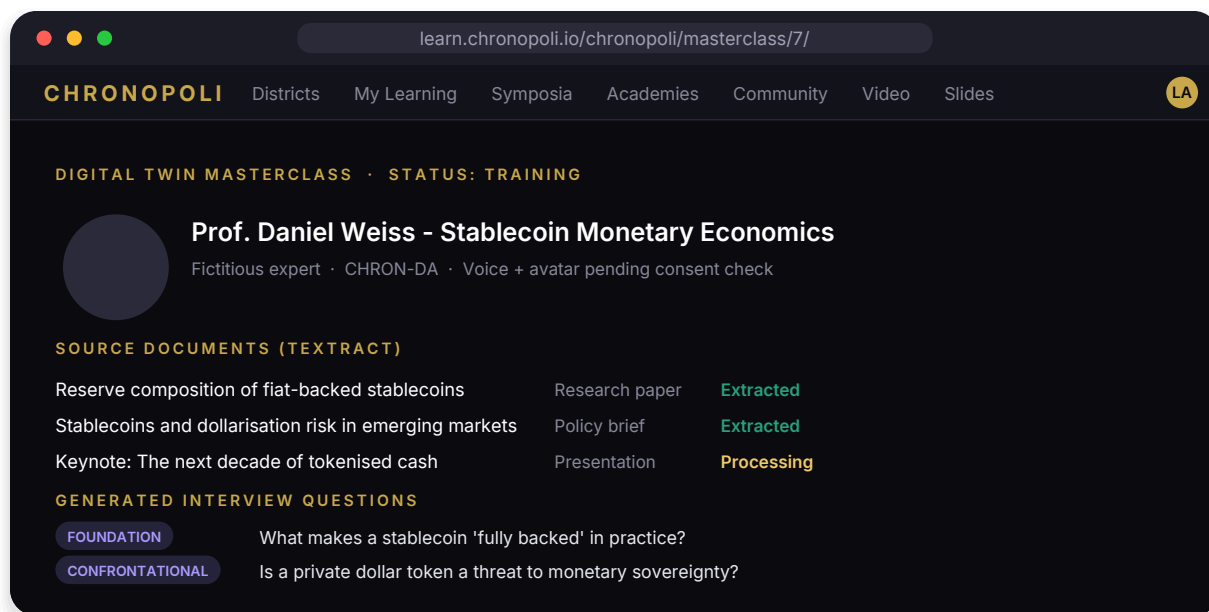


FIG. 8 A twin in training. Documents are extracted with Amazon Textract, and interview questions are generated per category.

STEP	WHAT HAPPENS	SERVICE USED
Upload sources	Research papers, policy briefs, book chapters and slides are uploaded	S3 + Amazon Textract
Analyse	Main arguments, positions and vocabulary are extracted into a profile	Amazon Bedrock (Claude)
Interview prep	Foundation, deep-dive, confrontational and audience questions are drafted	Amazon Bedrock (Claude)
Voice and avatar	Created only with the expert's written consent	ElevenLabs, HeyGen
Live session	Streamed like a round table and can be linked to a symposium	AWS IVS

● IMPORTANT

Consent comes first. A twin cannot be published without the expert's approval, and the expert can withdraw it at any time.

— CHAPTER 13

Administrator quick reference

Where to find things and how to run routine operations.

URL map

PATH	PURPOSE	AUDIENCE
/chronopoli/onboarding/	Five-question onboarding and results	Learners
/chronopoli/partners/	Partner directory and partner detail pages	Public
/chronopoli/partners/<slug>/dashboard/	Partner KPI dashboard	Partner staff
/chronopoli/ecommerce/checkout/<course_key>/	Stripe checkout for one course	Learners
/chronopoli/ecommerce/webhook/stripe/	Stripe webhook endpoint (signed)	Stripe
/chronopoli/academy/	Company Academy home, pathways, badges, jobs	Learners
/chronopoli/symposia/...	Round-table list, detail, schedule and approval	Members, moderators
/chronopoli/ai-tutor/chat/	AI Tutor chat API used by the widget	Learners
/chronopoli/masterclass/	Digital Twin list, detail and uploads	Experts, staff
/auth/discourse/sso/	Discourse SSO handshake	Community

Routine commands

OPERATIONS

```
# start / stop the platform (Phase 1, Docker on EC2)
tutor local start -d
tutor local logs -f lms
# Django shell inside the LMS container
tutor local do exec lms ./manage.py lms shell
# deploy, health check and backup
./scripts/deploy.sh production
./scripts/healthcheck.sh          # 8-category smoke test
./scripts/backup.sh              # databases + media to S3
```



PART FOUR

Technical review

Architecture, the open-source stack and its versions, 37 verified findings, the updates applied and the road to go-live.

IN THIS PART

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CHAPTER 14

Architecture overview

Chronopoli extends Open edX without forking it. A Tutor plugin injects settings, Django apps add the product features, and managed AWS services handle AI, media and the round-table pipeline.

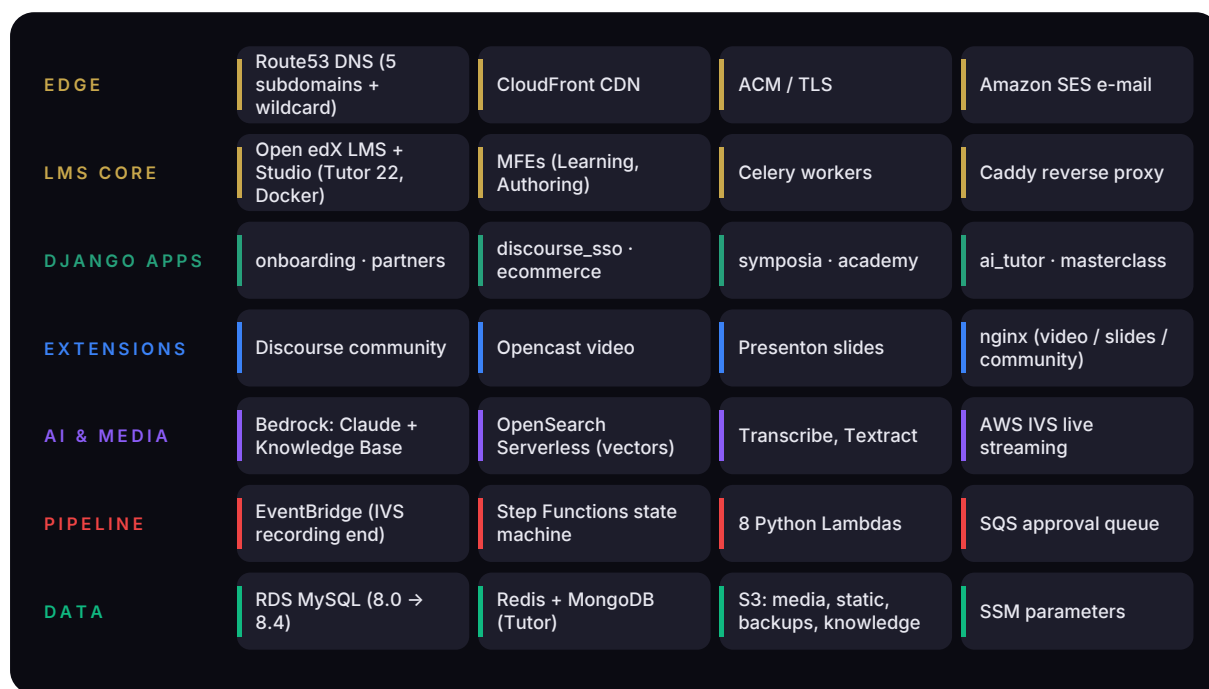


FIG. 9 Platform layers. Phase 1 runs on one EC2 host; Phase 2 moves the LMS core to EKS with autoscaling.

How the pieces connect

- **Tutor plugin** (tutorchronopoli): config defaults and generated secrets, settings patches for LMS and CMS, and an init task that creates the seven district organisations.
- **Django apps** hook into Open edX through signals in AppConfig.ready(): REGISTER_USER creates the onboarding profile, CourseEnrollment post_save updates purchases, and certificate events award badges.
- **Celery** runs symposia and masterclass work plus the weekly partner report.
- **Round-table pipeline**: IVS recording end, then EventBridge, then Step Functions (Transcribe, Claude analysis, four parallel generators, notify), then SQS, then the Django approval queue, then a task-token callback.
- **AI Tutor**: Bedrock knowledge-base retrieval plus a streaming Claude call, served over Server-Sent Events to the theme widget. The widget uses textContent only, and no XSS was found.

Review method

CHECK	RESULT
python -m py_compile on all 90 Python files	Pass
bash -n on every shell script and the Tutor init task	Pass
Django system check in a scratch harness (plugins linked as chronopoli_*)	Pass. Migrations match the schema; only help_text and label drift remain
URL resolution against the real urls.py files	3 routing defects confirmed (R-2, R-11, R-21)
Terraform reference scan (resources, vars, locals)	No undefined references. terraform validate not available offline
Dependency versions against PyPI, GitHub releases, ghcr.io and Tutor defaults	37 components inventoried (next chapter)

CHAPTER 15

Open-source stack and versions

Every third-party component Chronopoli uses, what it does for the platform, and how far it was from the current stable release when the review started.

COMPONENT	ROLE IN CHRONOPOLI	PINNED BEFORE	LATEST (SEP 2026)	LICENSE
Tutor	Deploys Open edX with Docker	<22 (plugin)	22.0.2 (Verawood)	AGPL-3.0
Open edX platform	LMS, Studio, MFES	Follows Tutor; docs say Redwood	release/verawood	AGPL-3.0
Django / Celery	Plugin framework, async tasks	From the Open edX image	Django 5.2 LTS / Celery 5.6	BSD-3
Terraform CLI	Infrastructure as code	>= 1.5.0	1.16.4	BUSL-1.1
AWS provider	All AWS resources	~> 5.0 (lock 5.100.0)	6.66.0	MPL-2.0
archive provider	Zips Lambda code	undeclared	2.8.1	MPL-2.0
RDS MySQL	Open edX database	8.0	8.4 LTS	GPL-2.0
Ubuntu (EC2 AMI)	Host OS	22.04 jammy	24.04 LTS noble	various
Docker + Compose	Runs Tutor and extensions	unpinned, two install paths	Ubuntu docker.io + compose-v2	Apache-2.0
AWS Lambda runtime	8 pipeline functions	python3.11	python3.13 / 3.14	-
Claude on Bedrock	Tutor, analysis, content generation	claude-3-5-sonnet (retired)	Sonnet 4.6 global / Sonnet 5.5	commercial
Titan Embeddings v2	Knowledge-base vectors	in me-central-1	use eu-central-1 / eu-west-1	commercial
boto3	AWS SDK	unpinned	1.43.104	Apache-2.0
stripe	Checkout + Connect	not installed	15.6.1	MIT
WeasyPrint / Jinja2	Opinion-paper PDF	unpinned	70.0 / 3.1.6	BSD-3
Opencast allinone	Video platform	latest	20.4	ECL-2.0
Presenton	AI slide builder	latest	v0.9.11-beta	Apache-2.0
Discourse	Community forum	tests-passed branch	latest (renamed)	GPL-2.0
actions/checkout	CI checkout	v4 (Node 20 removed)	v7.0.1	MIT
nginx	Proxy for extensions	Ubuntu package	Ubuntu package	BSD-2
Inter / JetBrains Mono	Theme typography	Google Fonts	-	OFL-1.1

● URGENT

Three items were urgent. **(1)** The Claude model ID used in 8 places is retired. **(2)** RDS MySQL 8.0 left standard support on 31 July 2026 and now incurs Extended Support fees. **(3)** The plugin's Tutor cap of <22 would have downgraded the platform to an end-of-life Open edX release.

CHAPTER 16

Review findings

37 verified findings, sorted by severity. Each one was confirmed by reading the code or by running a check. Findings marked FIXED were corrected, tested and pushed on the review branch in 15 commits.

Summary by severity						
6		18		13		8
CRITICAL		HIGH		MEDIUM		FIXED NOW
Block go-live		Fix before launch		Fix in first sprint		On the review branch
ID	SEVERITY	AREA	FINDING	FIX	STATUS	
R-1	CRITICAL	Packaging	The eight Django apps in plugins/ are never pip-installed. Folder names (ai-onboarding) do not match module names (chronopoli_onboarding), so LMS/CMS fail with ModuleNotFoundError.	Give each app a pyproject.toml and install it through OPENEDX_EXT_RA_PIP_REQUIREMENTS or a Dockerfile patch, then rebuild the image.	Open	
R-2	CRITICAL	Routing	URLs are added through a Tutor patch, openedx-lms-urls, that Tutor never renders, so no Chronopoli endpoint is routed.	Register the apps as edx-platform plugins (lms.djangoapp entry point + plugin_app url_config/settings_config).	Open	
R-3	CRITICAL	Pipeline	The CheckTranscription ResultSelector expects transcript keys that the Lambda returns only after COMPLETED. The first poll fails with States.Runtime.	Always return both keys, as empty strings until the job completes.	FIXED	
R-4	CRITICAL	Pipeline	The EventBridge input reads fields that the IVS event does not send. The recording prefix points at HLS segments, not an MP4, and round_table_id becomes the channel name.	Map recording_s3_bucket_name / key_prefix, render an MP4 (IVS rendition or MediaConvert), and carry the RoundTable ID in a channel tag.	Open	
R-5	CRITICAL	Pipeline	The approval hand-off is broken end to end: no waitForTaskToken, a dict vs list schema mismatch, no task_token stored, and the poller is never scheduled.	Use lambda:invoke.waitForTaskToken, agree one message schema, store the token and call send_approval from the approve view.	Open	
R-6	CRITICAL	Security	The media bucket is public-read and also holds grade CSVs (learner PII), course exports and expert documents.	Keep media private behind CloudFront OAC with signed URLs, and move grades and exports to a private bucket.	Open	

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-7	HIGH	Settings	getattr(globals(), ...) always returns {}, which wipes the Celery beat schedule. The referenced weekly-report task does not exist.	Fixed with globals().get() and a new send_all_weekly_reports task. Tutor still needs a beat process.	FIXED
R-8	HIGH	Tutor	The init task is never registered, and several scripts call manage.py without the lms argument. backup.sh always aborts.	Register it with CLI_DO_INIT_TASKS and use ./manage.py lms ...	FIXED
R-9	HIGH	Templates	Mako templates are rendered by the Django engine and show up as literal text. \${csrf_token} breaks onboarding CSRF.	Render them with edxmako render_to_response; make the Django templates standalone.	Open
R-10	HIGH	Templates	Templates are missing for ecommerce success/cancel/subscriptions, the partner dashboard and all masterclass pages, so those pages return 500.	Create the templates.	Open
R-11	HIGH	E-commerce	checkout/<course_key>/ is matched before checkout/success/, so Stripe's success_url returns 404.	Order the routes so success/ and cancel/ come first.	FIXED
R-12	HIGH	E-commerce	Payment is not enforced inside Open edX. Learners can enrol for free through the standard LMS button or API.	Use a paid CourseMode, or an openedx-filters enrolment pipeline step.	Open
R-13	HIGH	E-commerce	stripe_transfer_id is unique with a default of "", so only the first partner payout is ever saved. Status is never set to paid.	Make it nullable, store None, and update the status from Stripe webhooks.	PARTIAL
R-14	HIGH	E-commerce	The webhook marks purchases paid and enrolled without checking payment_status, and even when enrolment failed. Refunds are ignored.	Check payment_status == paid, return the real enrolment result, and handle async payments and refunds.	PARTIAL
R-15	HIGH	AI Tutor	The history window can start with an assistant turn, which Bedrock rejects. The tutor fails from about the 6th question on.	Trim the window to start at a user turn and merge consecutive turns from the same role.	FIXED
R-16	HIGH	IAM	bedrock-agent:Retrieve is not an IAM action, so knowledge-base retrieval is denied. StartIngestionJob is missing.	Use bedrock:Retrieve, bedrock:RetrieveAndGenerate and bedrock:StartIngestionJob.	FIXED
R-17	HIGH	IAM	The EC2 role lacks SES, SQS, IVS, Step Functions and S3 (knowledge/outputs) permissions that the Django apps need.	Add a least-privilege policy for each service.	Open

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-18	HIGH	Terraform	The Bedrock knowledge base has no OpenSearch data-access policy and no vector index, so apply fails. The network policy is public.	Add the access policy and create the index; restrict network access.	Open
R-19	HIGH	Terraform	The CloudFront origin is an IP address, which is not allowed. http-only plus Caddy HTTPS would cause a redirect loop.	Use a DNS origin (origin.learn...) with https-only.	Open
R-20	HIGH	Lambda	The opinion-paper Lambda imports jinja2 and weasyprint, but only handler.py is zipped, and WeasyPrint needs native libraries.	Ship it as a container-image Lambda (or a layer).	Open
R-21	HIGH	Symposia	The approval UI calls a URL that does not resolve, and routes are prefixed twice (/symposia/symposia/).	Use {% url %} and drop the doubled prefix.	FIXED
R-22	HIGH	Academy	The auto-badge signal imports CERTIFICATE_CREATED from a module that does not export it, so auto-badging is silently disabled.	Use openedx_events CERTIFICATE_CREATED and read certificate.user / course_key.	Open
R-23	HIGH	Academy	The subdomain middleware is never registered, matches slug rather than subdomain, and has no Caddy host, TLS or ALLOWED_HOSTS entry.	Register it, match on subdomain + academy_enabled, and add wildcard hosting.	Open
R-24	HIGH	Symposia	IVS channels are created without a recording configuration. create_channel and start_pipeline are never called, and the input keys do not match.	Pass recordingConfigurationArn, wire the calls and use unique execution names.	Open
R-25	MEDIUM	Privacy	The intelligence-reports API returns every partner's reports to any partner.	Filter by partner.	PARTIAL
R-26	MEDIUM	Auth	Dashboard access is granted when the user's e-mail equals the partner contact e-mail, and that e-mail is unverified.	Add an explicit partner-user relation.	Open
R-27	MEDIUM	Auth	Discourse SSO forwards unverified e-mails, which allows a takeover of an existing forum account.	Send require_activation for inactive accounts.	FIXED
R-28	MEDIUM	Config	The Discourse SSO secret is generated by Tutor, but setup-discourse.sh reads it from an environment variable. Nothing keeps the two in sync.	Read it with tutor config printvalue.	Open
R-29	MEDIUM	Privacy	Draft round tables and unapproved AI outputs are visible to any logged-in user.	Restrict them to staff and the initiator.	Open

ID	SEVERITY	AREA	FINDING	FIX	STATUS
R-30	MEDIUM	Masterclass	Unpublished twins are visible, and the upload view is a stub. Synchronous Textract fails on multi-page PDFs.	Add a publish check, a real upload and async Textract.	Open
R-31	MEDIUM	Settings	AWS_REGION, S3_MEDIA_BUCKET, SYMPOSIA_SFN_ARN and similar settings are read but never defined, and region defaults disagree.	Define them in the plugin settings patch.	Open
R-32	MEDIUM	AI Tutor	Bad JSON gives a 500. course_key goes unvalidated into the system prompt, and there is no rate limit on paid Bedrock calls.	Validate input and add a per-user quota.	Open
R-33	MEDIUM	Onboarding	Posted answers are not checked against the allowed options.	Validate against ONBOARDING_QUESTIONS.	Open
R-34	MEDIUM	Scripts	configure-tutor.sh can block on stdin, overwrites a tracked patch file and sets public-read ACLs.	Refactor the script alongside finding 6.	Open
R-35	MEDIUM	Terraform	SSM parameters default to empty strings, which SSM rejects.	Create them conditionally or make the variables required.	Open
R-36	MEDIUM	CI/CD	The deploy workflows restart Tutor but never run migrations.	Run tutor local do init --limit=chronopoli.	Open
R-37	MEDIUM	Privacy	Badge assertion IDs are inconsistent and recipient hashes are unsalted. The talent pipeline auto-applies learners without consent (PDPL risk).	Fix the assertion data and add an explicit opt-in.	Open

— CHAPTER 17

Updates applied and road to go-live

What changed on the review branch, and a six-sprint plan for the open findings.

Updates applied on branch `claude/zealous-knuth-y3ddq4`

AREA	BEFORE	AFTER
Tutor / Open edX	tutor>=18,<22 (Ulmo, EOL)	tutor>=22.0,<23 (Verawood); Python >= 3.10
Tutor install	unpinned tutor[full], system pip	tutor[full]>=22.0.2,<23 in /opt/chronopoli/venv
Terraform	>= 1.5.0; archive provider undeclared	>= 1.14.0; archive ~> 2.8 locked (validate passes on 1.16.4)
EC2 AMI	Ubuntu 22.04 jammy	Ubuntu 24.04 noble (gp3)
Docker packages	docker-compose-plugin (not in Ubuntu)	docker-compose-v2 + python3-venv
RDS MySQL	8.0 (extended support fees)	8.4 LTS
Lambda runtime	python3.11 (x8)	python3.13 (x8)
Lambda deps	unpinned; unused anthropic	boto3 1.43.104, weasyprint 70.0, jinja2 3.1.6
Claude on Bedrock	claude-3-5-sonnet-20241022 (retired), hardcoded x8	global.anthropic.claude-sonnet-4-6 via BEDROCK_MODEL_ID
Bedrock IAM	bedrock-agent:Retrieve (invalid)	bedrock:Retrieve / RetrieveAndGenerate / StartIngestionJob + global profiles
Opencast	allinone:latest, DB_VENDOR MySQL	allinone:20.4, MariaDB driver
Presenton	presenton:latest	presenton:v0.9.11-beta
Discourse	tests-passed branch	latest (renamed channel)
GitHub Actions	actions/checkout@v4 (Node 20)	actions/checkout@v7
Stripe SDK	not installed	stripe==15.6.1 via OPENEDX_EXTRA_PIP_REQUIREMENTS; webhook adapted to 15.x objects
Docs	Redwood, PostgreSQL 14	Verawood (Tutor 22), MySQL 8.4

● HELD BACK

Deliberately **not** changed yet: AWS provider 5.x to 6.x (a breaking major release that needs a reviewed terraform plan) and Claude Sonnet 5.5 (needs the temperature parameter removed from the Lambdas and a regional availability check). Both are planned for sprint 6.

Six-sprint plan to go-live

SPRINT	GOAL	FINDINGS
1 Boot	Package the Django apps as edx-platform plugins and fix routing, templates and init	R-1, R-2, R-9, R-10, R-31
2 Secure	Private media and exports, IAM least privilege, partner access model, consent	R-6, R-17, R-25, R-26, R-29, R-37
3 Commerce	Paid course modes, refunds, payout status, webhook hardening	R-12, R-14 (rest)
4 Pipeline	IVS recording, event mapping, task tokens, container Lambda for PDFs	R-4, R-5, R-20, R-24
5 AI & data	Knowledge-base index and access policy, region for embeddings, quotas	R-18, R-32, R-30
6 Launch	CloudFront DNS origin, wildcard academies, CI migrations, AWS provider 6.x	R-19, R-23, R-35, R-36

— CHAPTER 18

Glossary

Short definitions of the terms used in this booklet.

TERM	MEANING
District	An Open edX organisation (for example CHRON-COMP) that groups courses by field of knowledge
Layer	Depth of a course, from L1 Entry to L4 Influence
Company Academy	A partner's white-label space with pathways, badges and job postings on its own subdomain
Pathway	An ordered set of courses that ends with a badge
Open Badge	A verifiable digital credential following the IMS Open Badges 2.0 standard
Round table / Symposium	A closed, streamed expert session under the Chatham House Rule, with AI-generated outputs
Digital Twin	An expert-approved AI version of a person, trained on their published work
USDT	Tether's US-dollar stablecoin, used here only as an illustrative example
VASP	Virtual Asset Service Provider, such as an exchange or custodian
Travel Rule	FATF Recommendation 16: originator and beneficiary data must travel with a transfer
Tutor	The official Docker-based distribution and deployment tool for Open edX
Bedrock	Amazon's managed service for foundation models, used here to run Claude